

Coding & Solution

Date	01 July 2026
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

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The Voice-Based Concept Understanding Analyser is an AI-powered web application that evaluates a user's spoken explanation of a concept. The application converts speech into text using Whisper, analyzes conceptual understanding using NLP techniques, visualizes the uploaded audio waveform, and generates a detailed PDF report containing evaluation metrics. The solution is developed using Python and Streamlit with a modular architecture to ensure readability, maintainability, and future scalability.

Solution Summary

Field	Details
Repository Link / URL	Paste your GitHub Repository URL (e.g., https://github.com/username/Voice-Based-Concept-Understanding-Analyser)
Programming Language(s)	Python
Framework(s) Used	Streamlit, Whisper, Sentence-BERT, Librosa, ReportLab
Key Features Implemented	• Speech-to-Text Conversion using Whisper • Voice Upload Interface • Audio Waveform Visualization • Concept Understanding Analysis • Confidence Score Generation • PDF Report Generation • Interactive Streamlit Dashboard
Pending / Incomplete Features	Real-time microphone recording, User Login, Database integration, Advanced analytics dashboard (if not completed). If everything is finished, write None.
Setup / Run Instructions	1. Install Python 3.10+ 2. Install dependencies using <code>pip install -r requirements.txt</code> 3. Run the application using <code>streamlit run app.py</code> 4. Upload a .wav or .mp3 audio file and view the generated report.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (If uploaded to GitHub; otherwise write No.)

Additional Notes / Comments:

The project follows a modular software architecture with separate components for speech recognition, concept analysis, audio visualization, report generation, and the Streamlit user interface. The code is readable, reusable, and maintainable by following Python coding standards, proper naming conventions, comments, and error handling. The solution can be extended in the future by adding multilingual speech support, real-time microphone input, cloud deployment, and user authentication.